AMINO ACIDS AND PROTEINS 6

Proteins are needed in the body for the growth and repair of tissues and to help certain chemical reactions take place. They are used as an energy source only under extreme conditions. (Energy is the capacity to do work.)

1 Name one food group that is a major source of energy.

Proteins are very complex molecules. They are organic compounds that are used to make enzymes (biological catalysts) and hormones in the body. A hormone is a chemical produced by a gland and then carried in the bloodstream to regulate body functions. Hormones act as chemical messengers sent to produce a specific reaction. Proteins are **polymers** (very large molecules) that are made up of smaller molecules called **amino acids**. Amino acids contain both the $-NH_2$ (amine) and -COOH (carboxylic acid) groups.

Natural amino acids have quite complex structures so are often referred to by their 'trivial' names rather than by IUPAC nomenclature. They can be represented by the general formula $H_2N–CH(Z)–COOH$. (See figure 1.) The 'Z' represents the side chain and is responsible for the differences between amino acids. In α-amino acids, the carboxyl, amino and side chain are all attached to one central carbon atom. In aqueous solution the amino group can act as a base and the carboxyl group as an acid.

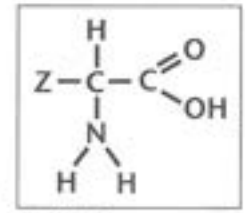


Figure 1

2 The carboxyl and amino groups are very polar. How will this affect the solubility of amino acids in water?

3 Outline how amino acids were named.

4 Glycine is the simplest amino acid. Its side chain ('Z') is hydrogen. Draw the structural formula of glycine.

5 Alanine has the side chain $–CH_3$. Draw the structural formula of alanine.

Amino acids can join together to form longer molecules. If the chain is up to 10 amino acids long, it is sometimes called a **peptide**.

6 What do you think a *polypeptide* is?

In living things there are about 20 different α-amino acids. Proteins make up about 50% of a human body's dry weight. They are found in all body cells and have these functions:

- Make up muscles and tendons
- Give the skin its elasticity and strength
- Reinforce bones and teeth
- Make up antibodies, which are part of the body's immune system
- Make up haemoglobin, which is part of the blood's transport system
- Make up enzymes and hormones, which control chemical reactions in the body.

The name **protein** is derived from the Greek *proteios*, meaning 'primary' or 'of the first rank'.

7 Why do children need a greater percentage of proteins in their diet than adults do?

Proteins are formed when amino acids join together by condensation reactions. In these reactions, a carbonyl to nitrogen bond is created. An –OH group from one molecule and a –H from another molecule form a separate molecule of water. This is called a **peptide bond**.

8 Is this carbon to nitrogen bond covalent or ionic? Give a reason for your answer.

Figure 2 shows how two simple amino acids, glycine and alanine, can react to form a dipeptide.

9 Show another way in which glycine and alanine can join. (Hint: reverse the positions of the two amino acids as shown in figure 2).

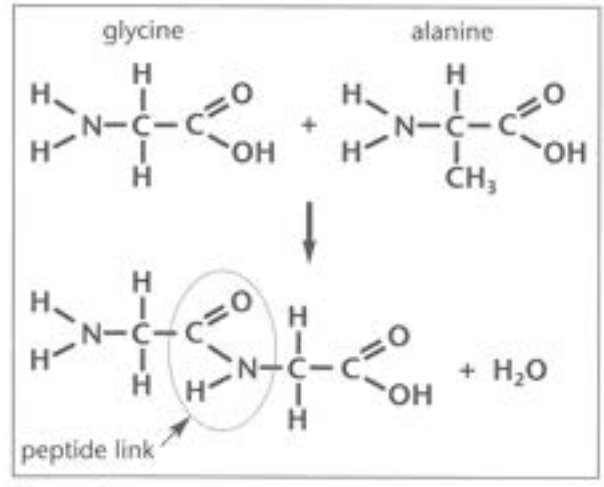


Figure 2

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Protein structure

How proteins behave and what they do depends upon their shape. Proteins have **primary**, **secondary** and **tertiary** structures.

Primary structure is the order in which the amino acids are arranged; e.g. alanine, followed by glycine then lysine (ala - gly - lys) and so on. In other words, primary structure is the building of the initial chain.

Secondary structure is when the chain coils or folds to form a three-dimensional shape. This is because parts of the chain can attract each other. The two most common shapes are a flat sheet and a helix (helical). In other words, the secondary structure is the twisting of the chain that was formed into its given shape.

10 Draw the shape of a helix. (You may need to refer to a textbook.)

The attractions between the parts of the protein chain are mainly due to hydrogen bonds between –C=O and $-NH_2$ groups in neighbouring amino acids.

11 What is a –C=O group called?

12 What is a $-NH_2$ group called?

Hydrogen bonds are the weak attractions between dipoles (slightly-charged areas) existing in polar molecules. Figure 3 shows the hydrogen bonding between oxygen (δ–, slightly negative) and hydrogen (δ+, slightly positive).

13 Copy figure 3 and show the relative charges on the atoms.

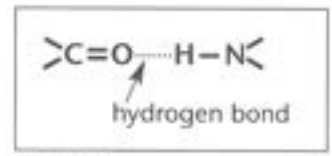


Figure 3

Tertiary structure is when distortions occur in the three-dimensional shape of a protein. These distortions are caused by the large Z-groups of some amino acids. There can also be hydrogen bonding between positive ($-NH_3^+$) and negative ($-COO^-$) groups. In other words, the tertiary structure is a further shaping of the secondary structure.

14 Outline the differences between the primary, secondary and tertiary structures of proteins.

The formation of a peptide link between amino acids can be reversed. Proteins can be hydrolysed (broken down) by heating with aqueous acids. In digestion, proteins are hydrolysed much more easily by using **digestive enzymes** such as pepsin and trypsin. **Hydrolysis** (reaction with water/OH^-/H^+) can be used to identify the amino acids present in a protein. A protein can be hydrolysed and the products used in paper chromatography to separate and identify the components.

15 Copy figure 4 and show the products of this reaction.

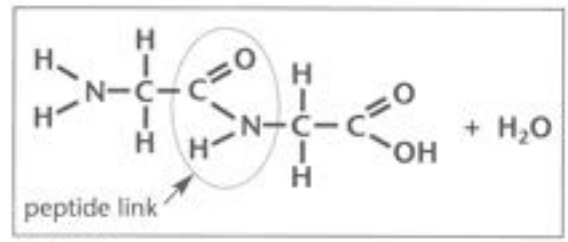
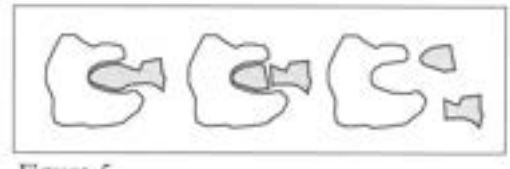


Figure 4

Enzymes Enzymes are proteins that control the reactions which take place in the body. Enzymes have some similarities with inorganic catalysts (i.e. they speed up reactions) but there are important differences. They are more specific than inorganic catalysts. Enzymes do not catalyse lots of reactions, but usually specialise in producing one reaction only. Enzymes work under milder conditions and tend to operate at body temperatures instead of the high temperatures and pressures needed in many industrial processes.

The secondary and tertiary structures give each enzyme a unique cavity or pocket called an **active site**. This active site will only hold molecules of a specific shape; this is why it is often called a **lock-and-key** structure. The sequence of an enzyme-catalysed reaction is:

- The reactant (substrate) goes into the active site
- Bonds are formed between the substrate and the enzyme
- Bonds are broken within the substrate
- The substrate breaks or rearranges into new products
- The products are released from the active site.


Figure 5

16 Copy figure 5 and label the following: *enzyme, active site, substrate, product.*

17 Outline the similarities and differences between inorganic catalysts and enzymes.

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